DSA REFERENCE

Graph Shortest Path

Recognition cues, invariants, code templates, and 5-8 referenced problems.

Expand paths in an order that guarantees a finalized distance cannot later improve.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

DSA / SHORTEST-PATH

DSA

Graph Shortest Path

CORE IDEA

Expand paths in an order that guarantees a finalized distance cannot later improve.

Recognition cues

+ minimum weighted cost
+ nonnegative edges
+ few negative edges requiring another algorithm
+ path reconstruction

Implementation sequence

+ classify edge weights
+ choose BFS, 0-1 BFS, Dijkstra, or Bellman-Ford
+ relax edges
+ skip stale heap entries
+ store parents if needed

Invariant

A popped non-stale Dijkstra entry has the smallest unsettled distance; finalized distances are optimal.

Worked reasoning

Dijkstra pushes a new pair whenever dist[u] + weight improves dist[v]; stale older pairs are ignored on pop.

Complexity

Dijkstra with adjacency lists and a binary heap is O((V + E) log V); storage is O(V + E).

Failure modes

- [ ] using Dijkstra with negative edges
- [ ] marking visited on push
- [ ] overflow in large distances
- [ ] forgetting unreachable nodes

Practice ladder

Graph Shortest Path: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

#	Problem	Deliberate-practice prompt
1	**Network Delay Time**	State the cue, invariant, edge case, and proof before implementation.
2	**Cheapest Flights Within K Stops**	State the cue, invariant, edge case, and proof before implementation.
3	**Path With Minimum Effort**	State the cue, invariant, edge case, and proof before implementation.
4	**Swim in Rising Water**	State the cue, invariant, edge case, and proof before implementation.
5	**Find the City**	State the cue, invariant, edge case, and proof before implementation.

Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

PUBLISHING STATUS

REVIEW GATE: STABLE TECHNICAL. AI-produced draft. A named human must verify technical accuracy, sourcing, freshness, and editorial quality before publication. Reviewed-through date: 2026-07-18.